

Provenance-Certified KKR Convergence Assistance: ml_assist, PotentialBank v1, and Same-Structure Warm Starts

An AiiDA-native early-abort framework and a certified potential-reuse bank —
with a live pilot, a held-out benchmark, and explicitly benchmarked limits

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Abstract

Self-consistent-field (SCF) convergence is the practical bottleneck of KKR and Bogoliubov-de Gennes (BdG) Green’s-function DFT campaigns: many runs stall or diverge, wasting cluster time. We present an **AiiDA-native convergence-control framework**, `ml_assist`, that scores a run’s first ten SCF iterations and can abort doomed runs early — an opt-in layer with a zero-risk default. The framework ships a **frozen numpy random-forest scorer** (the AiiDA daemon has no scikit-learn) alongside a **simple RMS-at-iteration-10 baseline**, both wrapped in a **per-cohort conformal early-abort policy** that controls the false-abort rate after per-cohort calibration under calibration/test exchangeability, with Mondrian per-stratum thresholds for conditional coverage. A **live Fe-doped NbSe₂ pilot validates the abort machinery end-to-end**: a true early abort saved ~ 130 SCF iterations against its diverging counterfactual, and a would-be false abort on a slowly-converging run was correctly withheld ($N=5$, descriptive). On a **336-run, leakage-clean held-out replay with per-run convergence-target (QBOUND) labels**, the learned model does *not* beat the RMS@10 baseline on the primary NbSe₂ normal-state cohort (held-out AUC 0.83 vs. 0.90), *does not transfer across materials* (other-material AUC $\approx$ chance), and a NbSe₂-seeded conformal threshold does not transport (realized false-abort 0.47) — though **per-cohort recalibration restores** realized $\approx$ target α . We conclude that *reliable deployment requires per-cohort calibration and strong simple baselines*; the honest, provenance-tracked framework — not model superiority — is the contribution. We then present **PotentialBank v1**, a provenance-certified extension of AiiDA-KKR that realizes the warm-start lever: it retrieves, certifies, and reuses compatible *same-structure* converged potentials. AiiDA-KKR already provides the restart and `potential_overwrite` primitives; our contribution is the **certification and retrieval layer above them** — byte-exact potential integrity (three-way SHA-256), exact Z -sequence/NATYP compatibility, non-CPA/non-BdG scope control, and no guessed mapping. PotentialBank v1 is frozen at **237 unique certified donors (238 nodes)** across **8 material families** and **33 F2-eligible materials**, with same-structure warm starts cutting SCF iterations substantially (per-family, mixing- and contour-qualified). We claim *no cross-material transfer and no guaranteed acceleration*: donors reuse the identical structure, and nine hard cases are documented with root causes. Separately, an engineering effort repaired four cell-inappropriate BdG-workchain defaults, enabling the first end-to-end multi-atom BdG run and its memory-scaling number. Everything ships as an installable `aiida-kr-mlassist` extension (with an optional convergence advisor), unit tests, and pre-registered success gates.

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Keywords. KKR Green-function DFT; SCF convergence; AiiDA provenance; potential reuse / warm start; conformal prediction; reproducible workflows; certified donor potentials.

Program summary

Program title: `aiida-kkr-mlassist` / KKR Convergence Advisor / PotentialBank v1.

Developer’s repository: https://github.com/MRHemmati/kkr_ml_tools (CPC library link and DOI assigned on acceptance).

Licensing provisions: MIT (open source).

Programming language: Python (pure-`numpy` runtime core; AiiDA / `aiida-kkr` used only for submission).

Nature of problem: KKR SCF convergence wastes HPC time when runs stall or diverge; restarting from a converged potential is standard and AiiDA-KKR provides the primitives, but choosing a *compatible*, *provenance-safe* donor is error-prone and unaudited.

Solution method: an AiiDA-native, opt-in `ml_assist` early-abort wrapper (pure-`numpy` scorer + RMS@10 baseline under a per-cohort conformal policy) and **PotentialBank v1**, a provenance-graph donor catalog admitting potentials only under strict E1 certification — byte-exact SHA-256 integrity, exact Z-sequence / NATYP compatibility, non-CPA / non-BdG scope, and no guessed mapping — with an optional same-structure warm-start advisor.

Restrictions: normal-state, non-CPA for PotentialBank v1; CPA / BdG and cross-material prediction are not supported; warm-start reuse is same-structure only, with no guaranteed acceleration; raw potentials remain in the AiiDA repository (referenced by SHA-256 / provenance).

Unusual features: strict provenance certification with byte-exact hashing, a no-guessed-mapping rule, per-family same-structure warm-start benchmarks, and a documented catalog of hard cases.

Additional comments: PotentialBank v1 is frozen at 237 unique certified donors (238 nodes) across 8 families, 33 F2-eligible materials.

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1 Goals and scope

The parent project is a multi-year KKR / KKR-BdG superconductivity effort on 2H-NbSe₂ with transition-metal dopants (V/Mn/Fe/Co), plus a Bi/NbBi subset. SCF convergence is the practical bottleneck: many runs stall, diverge, or converge slowly, wasting cluster time. `ml_assist` targets three *acceleration levers*:

L1 — early abort stop runs that are predicted not to converge, recovering their wasted tail.

L2 — probe-and-commit mixing race a few mixing settings briefly, commit the best.

L3 — warm start reuse a converged potential to halve iterations.

This report is in three parts. **Part 1** (Secs. 1–8) covers `ml_assist` — the early-abort framework (L1), its conformal policy, the live pilot, and the leakage-clean held-out replay. **Part 2** (Sec. 15) presents **PotentialBank v1**, the realized warm-start lever (L3): a certified catalog of single-structure converged potentials with an optional convergence advisor and measured same-structure savings. **Part 3** (Sec. 16 onward) gives limitations and roadmap. The probe-and-commit mixing search (L2) reuses the same infrastructure and remains estimated by the pre-registered campaigns.

Positioning. KKR restart and potential reuse are *not new*: AiiDA-KKR [1] already exposes the low-level primitives — a KKR calculation can continue from a parent remote folder, automatically extracting the converged potential, and the Voronoi step accepts an explicit `potential_overwrite`. Our contribution is the **layer above** these primitives. We introduce a provenance-certified extension of AiiDA-KKR that systematically retrieves, validates, and reuses compatible **same-structure** donor potentials for reproducible KKR warm starts, supplemented by an optional convergence advisor whose limits are explicitly benchmarked. Concretely: a provenance-graph procedure discovers candidate donors; each is admitted only under a strict eligibility policy (byte-exact potential integrity via a three-way SHA-256 match, exact *Z*-sequence / NATYP compatibility, non-CPA and non-BdG scope control, and no guessed mapping); and the resulting same-structure warm starts are benchmarked, with hard cases and negative results reported honestly. This turns an available mechanism into an auditable, policy-controlled reuse system.

Design principles. (1) *Opt-in, zero-risk default*: absent an `ml_assist` input the workflow behaves identically to stock. (2) *Advisory by default*; abort only when explicitly enabled and the statistical guarantee is certifiable. (3) *Per-system deployment*: we do not claim cross-material transfer. (4) *Honesty first*: negative results are reported as-is and are publishable.

2 Background and tooling (what we used and why)

JuKKR (KKRhost, voronoi, KKRimp). The Green’s-function DFT engine. SCF proceeds by iterating the potential; each iteration reports an RMS error and charge-neutrality. *Why it matters for ML*: these per-iteration quantities are the raw signal from which convergence fate is predicted.

BdG extension. Superconducting pairing is treated by doubling the Hamiltonian into Nambu (particle-hole) space; KKRhost-BdG additionally prints a pairing amplitude Δ per iteration. *Why*: the pairing channel is a BdG-specific failure mode invisible to normal-state features.

AiiDA 2.7. Workflow engine and provenance store. All calculations, inputs, and outputs are nodes in a directed provenance graph. *Why*: it gave us a mineable corpus of historical runs *and* is the natural place to deploy an opt-in wrapper workflow with full audit trail.

aiida-kk. The AiiDA plugin providing `kk_scf_wc` (normal SCF) and `kk_bdg_wc` (BdG chain). *Why*: `ml_assist` subclasses `kk_scf_wc` to add the prediction hook.

scikit-learn (training only). Random forests for the classifier. *Why RF*: in distribution it beats logistic regression and the physical extrapolation heuristic and needs no feature scaling (Table 3). *But out-of-sample* the strong baseline to beat is a plain RMS-at-iteration-10 rule, which the model does *not* beat on the primary cohort (Sec. 5). *Constraint*: scikit-learn is *not* installed in the AiiDA daemon environment.

numpy (inference). We export the trained forest to a flattened-array `.npz` and traverse it in pure `numpy`, reproducing scikit-learn’s `predict_proba` exactly. *Why*: the daemon must score runs without scikit-learn; a `numpy`-only path removes that dependency (Sec. 4).

Conformal prediction. A distribution-free wrapper that turns a score into a decision with a controlled error rate. *Why*: it lets us *control the false-abort rate* $\leq \alpha$ after per-cohort calibration under calibration/test exchangeability, regardless of the model’s calibration — the safety property an abort policy needs (Sec. 6).

3 Data and the provenance-mining audit

The AiiDA database holds $\sim 7,700$ calculation nodes and $\sim 6,400$ workflows (2021–2026). We mined per-iteration RMS/charge-neutrality trajectories and outcomes. Three *leakage-class* corrections were essential and define the trustworthiness of every downstream number:

1. **Segment de-inflation.** Training rows were KKR *segments* (up to 5 per workflow), not runs. Deduplicating to one cold-start row per run reduced 3,405 segments to **2,710 runs** (1,851 normal, 90% converged).¹

¹Modeling uses the 1,830 normal runs with at least $T_{\text{obs}}+2$ iterations (21 too-short runs excluded, needed to form the observation window); the BdG Δ study uses the 641 of 846 BdG runs whose `out_kkr` was retrievable and parseable (Sec. 9).

2. **Censoring scope.** 389 excepted/killed calculations are 100% censored; the dataset is *completed runs only*. Crashes/NaN-kills are out of scope and handled by a watchdog, not the model.
3. **The Replay-OOF Invariant.** Any replay, preview, or “deployment simulation” number is out-of-sample or prospective *by construction*, never in-sample. This was adopted after an in-sample shadow replay produced a spurious recall of 1.000; the honest out-of-sample number is 0.70 (Sec. 7). The model that scores a run never trained on it (verified by node id).

These corrections deflated some earlier claims (e.g. a cross-material transfer number collapsed from 0.94 to chance) — an outcome we treat as a contribution, not an embarrassment.

4 The convergence classifier

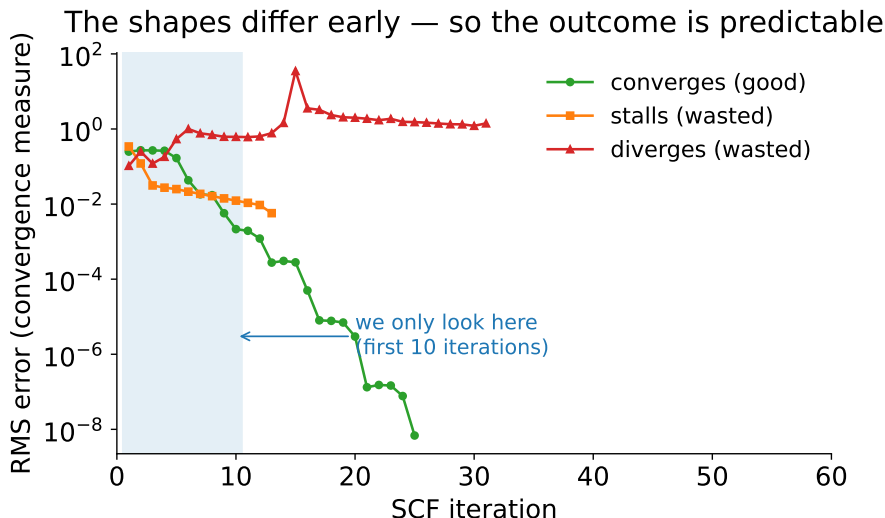


Figure 1: Why early prediction is possible. Converging, stalling, and diverging runs already trace distinct shapes within the first ten iterations (shaded) — the only data the model sees.

Features. A 15-dimensional vector computed from the first $T_{\text{obs}} = 10$ iterations: log-RMS window statistics (first, last, mean, std, min, drop, slope), two charge-neutrality statistics, and three mixing parameters each with a missing-flag. A single canonical implementation is shared by training and inference so they cannot drift (verified bit-identical).

Model and in-distribution accuracy. A 400-tree balanced random forest. Its *in-distribution* 5-fold cross-validation AUC is strong where data is plentiful and honestly weaker where it is thin (Table 2); the pooled normal-state 5-fold AUC is 0.944. In distribution it beats logistic regression and a physics extrapolation heuristic (Fig. 2, Table 3). **We deliberately do *not* treat 0.944 as the deployment headline:** it is a same-distribution cross-validation number. The deployment-relevant question — how the frozen model scores runs it has never seen, against a simple RMS baseline — is answered by the leakage-clean held-out benchmark of Sec. 5, and there the model does *not* beat RMS@10 and does *not* transfer across materials. The model is therefore deployed per-system with per-cohort calibration.

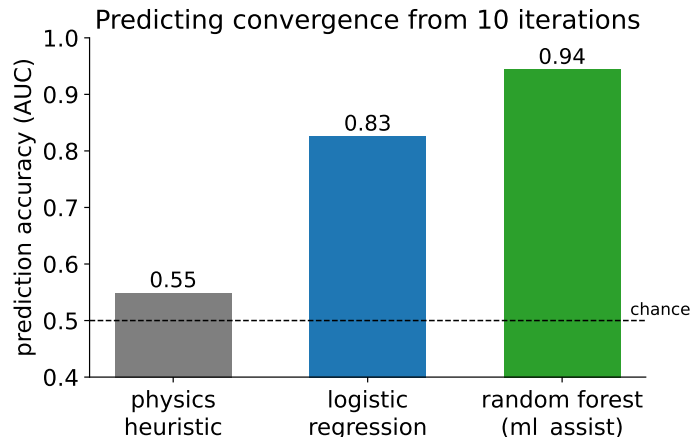


Figure 2: *In-distribution only*. From ten iterations, the random forest cross-validates better than a physics extrapolation heuristic or logistic regression (AUC; 0.5 is a coin flip). This is *not* the deployment metric — see the held-out benchmark (Fig. 3), where a simple RMS@10 rule matches or beats the model out-of-sample.

A dropped complication (reported for the record). A schedule-aligned observation window was a pre-registered design lock intended to protect “flat-start” runs (flat, high-RMS early, then converging). On the deduplicated data it gave *no* measurable benefit: the fixed window already separates flat-starters at AUC ~ 0.90 (the discriminative signal is in charge-neutrality and mixing, not the flat RMS slope). It was dropped for v1; the frozen model uses the plain fixed window.

Pure-numpy export and the float32 gotcha. The forest is flattened to node arrays (children, feature, threshold, per-leaf class probabilities) and traversed in `numpy`. Golden-parity against scikit-learn is **exactly** 0.0, verified running under the daemon interpreter (`numpy 1.26`, no `scikit-learn`). One subtlety was decisive: `scikit-learn` casts inputs to `float32` for tree traversal, so a `float64` comparison mis-routes samples sitting on a threshold tie (an initial 10^{-2} parity failure). The `numpy` path compares in `float32`; embedded golden vectors (including the exact edge sample that exposed the bug) guard against regression.

5 The held-out benchmark: does the model beat a simple RMS rule?

The honest test of deployment value is a *leakage-clean, out-of-sample* evaluation with correct labels. We assembled **336 completed ifflurm runs verified absent from training** (by node id) and scored each run’s frozen first-ten-iteration features. Three corrections over an earlier pooled analysis were decisive (they revise our own earlier, over-optimistic numbers):

1. **Per-run QBOUND labels.** Convergence is judged against each run’s own convergence target, not a fixed 10^{-3} ; a fixed cut mislabelled 57 runs (many NbSe2 runs use `QBOUND=0.008`).
2. **Censoring.** 15 excepted/killed runs (`exit` $\neq 0$) are dropped from all denominators.
3. **No pooled headline.** Numbers are reported *per cohort*; a single pooled AUC conflates a warm/restart population (which mostly converges) with cold starts (which mostly fail) and is not a deployment metric.

The result (Table 1, Fig. 3). On the primary target — NbSe₂-family normal-state — the frozen model reaches held-out AUC **0.83** [0.77, 0.89] (cold starts 0.79), while a **conformally-calibrated RMS@10 rule reaches 0.90** [0.85, 0.94] and gives *higher* failed-run recall and compute savings at matched false-abort at every α (Sec. 7). **The learned model does not beat the simple RMS baseline on the primary cohort.** It is anti-predictive across materials (other-KKR AUC 0.48; a warm/restart cohort 0.35) — a genuine transfer failure, since RMS@10 remains predictive on the same runs (0.90, 0.61). The one cohort where the model edges RMS@10 (“unknown”, 0.80 vs. 0.78) is small ($n=42$). Excluding the 56 borderline runs (final RMS within a factor 3 of QBOUND) leaves the ranking unchanged.

Table 1: Held-out benchmark, per cohort (leakage-clean, per-run QBOUND labels, censored dropped). AUC [95% bootstrap CI]. *No pooled headline number is reported.* A conformally-calibrated RMS@10 rule matches or beats the frozen model everywhere except the small “unknown” cohort; the model is anti-predictive across materials (other-KKR, warm/restart).

cohort	n	conv.	frozen ML	RMS@10	RMS-slope
NbSe₂-normal (primary, all)	167	38	0.831 [.77,.89]	0.897 [.85,.94]	0.740
— cold starts	128	12	0.785 [.68,.88]	0.901 [.80,.97]	0.568
— warm/restart	39	26	0.430	0.414	0.467
Bi/NbBi	28	21	0.898	0.939	0.816
other-KKR (cross-material)	45	34	0.476	0.904	0.583
BdG-misfit (normal model on BdG)	78	41	0.581	0.813	0.631
unknown	42	10	0.800	0.778	0.878
warm/restart-like (all families)	93	67	0.350	0.606	0.738

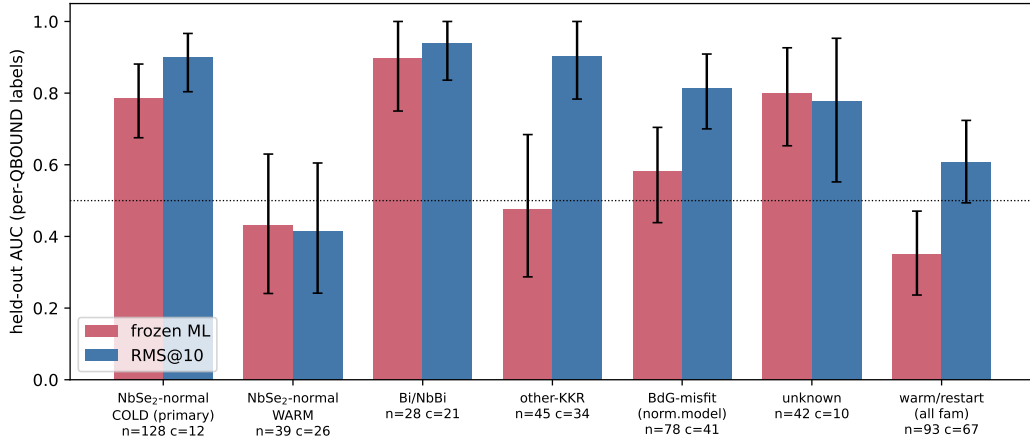


Figure 3: Held-out per-cohort AUC (per-run QBOUND labels), frozen ML vs. RMS@10, with 95% bootstrap CIs. On the deployment target (NbSe₂-normal cold, leftmost) RMS@10 leads; the model is at or below chance on other-material and warm/restart cohorts. There is no single pooled headline number.

Scope consequences. (i) BdG runs must be scored with the BdG model, not the normal one (normal-model-on-BdG AUC 0.58). (ii) Warm/restart runs are *outside the current abort policy*:

their failures are rare and not separable from the early window (both methods $\approx$ chance), so they are excluded from abort and left to run. (iii) No cross-material transfer is claimed for the learned model.

6 The conformal early-abort policy

Guarantee. Given a target α , we abort a run iff its predicted convergence probability is below a threshold calibrated on the campaign’s converged runs. One-sided split conformal makes

$$\Pr(\text{abort} \mid \text{the run would actually converge}) \leq \alpha.$$

Certiability needs ~ 19 calibration points at $\alpha = 0.05$ (~ 9 at 0.10); a campaign seeds these from material-matched history and refreshes online. The guarantee holds under *exchangeability* of calibration and test runs; material-matched seeding plus online refresh is how we keep that assumption approximately valid as a campaign drifts; a pre-registered violation trip-wire reverts to advisory mode if the realized false-abort rate ever exceeds the budget.

The marginal-vs-conditional gap (found and fixed). The split-conformal guarantee is *marginal*. Verified in out-of-sample replay: a single (marginal) threshold gives overall false-abort 0.049 at $\alpha = 0.05$ but **0.086 conditional on the flat-starter stratum** — the very runs the dropped aligned window was meant to protect. The fix is **Mondrian (group-conditional) thresholds**: one threshold per stratum (flat-starter vs. other, derived directly from the feature vector, agreeing 100% with the trajectory definition). This restores conditional control to **0.047** (Fig. 4, Table 4). Both vehicles deploy Mondrian.

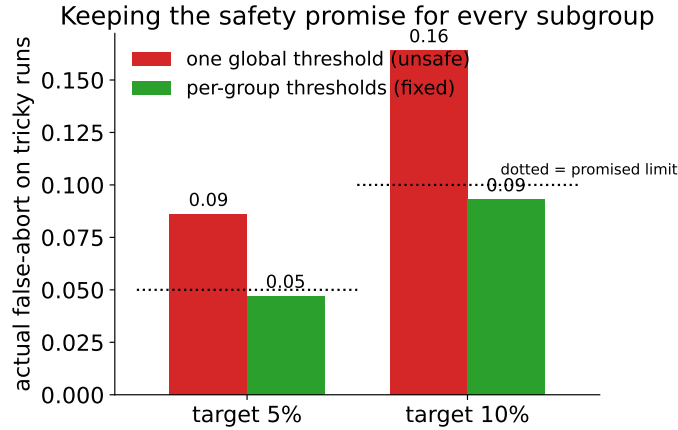


Figure 4: The safety fix. A single global threshold breaks the false-abort promise on the tricky “flat-start” runs (red); one threshold per subgroup (green) restores the target rate within the evaluated strata/cohort.

Seeded thresholds do not transport — recalibrate per cohort. The guarantee holds only under *exchangeability* of calibration and test runs, and the held-out data show this is not automatic. A threshold seeded on historical NbSe₂ converged scores, applied to the broader held-out cohort, gave a realized false-abort rate of **0.47** at $\alpha = 0.10$ — far above target: the seed does *not* transport across cohorts. The fix is **per-cohort recalibration**: re-deriving the threshold on the deployment

cohort’s own converged runs restores realized false-abort to $\approx$ target ($0.10 \rightarrow 0.098$ for the model, 0.100 for RMS@10; Fig. 5, nested-bootstrap split-conformal, model weights frozen). The conformal guarantee is thus a *per-cohort calibration* property, not a transported one; deployment must recalibrate on the target cohort, and the trip-wire reverts to advisory if the realized rate exceeds budget.

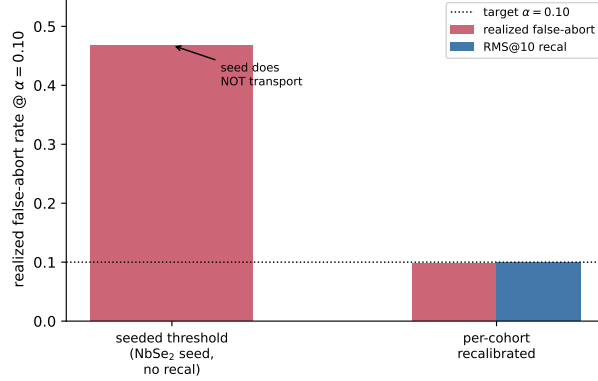


Figure 5: The conformal threshold is a per-cohort object. A NbSe₂-seeded threshold applied to held-out data breaks the false-abort promise (0.47 at $\alpha = 0.10$); per-cohort recalibration (weights unchanged) restores realized $\approx$ target for both the model and RMS@10.

Fallback discipline. For BdG, if the Δ channel is unparseable the policy falls back to an rms-only model *with its own* conformal threshold (never comparing an rms-only score to an rms+ Δ threshold); both thresholds are certified independently.

7 Deployment evaluation (per-cohort, calibrated)

All numbers are out-of-sample (Replay-OOF Invariant), accounted in SCF iterations (hardware-invariant), and evaluated *on the primary NbSe₂-normal cohort with per-cohort recalibration* (Sec. 5). Each abort is charged at the deployment horizon (cumulative iteration 20, when the model first scores under `nsteps=10`).

Recovery and false-abort, ML vs. RMS@10 (Fig. 6, Table 5). With per-cohort recalibration both policies hold realized false-abort at target ($\alpha = 0.05/0.10/0.20 \rightarrow 0.04/0.10/0.20$). At matched α the **RMS@10 baseline recovers more wasted compute and catches more doomed runs than the model**: failed-run recall 0.64/0.78/0.86 (RMS@10) vs. 0.53/0.62/0.71 (ML), and fraction of wasted iterations saved 0.61/0.73/0.81 vs. 0.54/0.62/0.69. The model’s abort precision stays high (≥ 0.96), but it is the weaker recovery policy on the primary cohort. This is the deployment consequence of the held-out benchmark: *a simple RMS@10 rule, per-cohort calibrated, is the stronger early-abort policy here.*

α economics. Because a false abort under the auto-retry ladder *warm-restarts* (it does not discard a converging run), its cost is small, which pushes the net-optimal α upward. The framework ships two profiles — $\alpha = 0.05$ (conservative) and $\alpha = 0.10$ (throughput) — but the choice of *scorer*

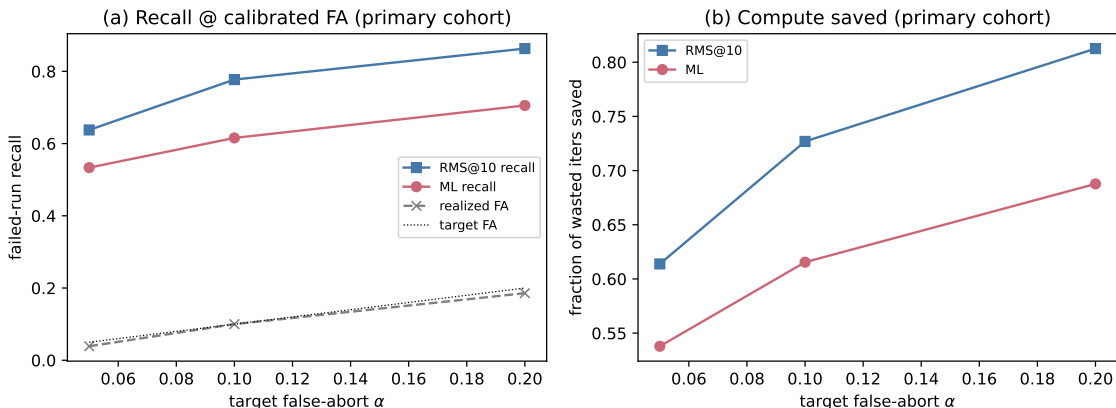


Figure 6: Calibrated risk-savings on the primary NbSe₂-normal cohort (per-cohort recalibration, weights frozen). (a) failed-run recall and (b) fraction of wasted iterations saved, both at matched realized false-abort. RMS@10 (blue) dominates the frozen model (red) at every α .

(RMS@10 vs. model) is now itself a per-cohort decision, with RMS@10 the default on the primary cohort.

The single-material prospective pilot (narrow). An earlier prospective check calibrated on 329 historical NbSe₂ converged scores and applied to a held-out 2026 NbSe₂ grid (20 runs, verified absent from training) gave a realized false-abort rate of 0.000 at $\alpha = 0.05$ and 0.10. **This is a narrow, single-material, in-distribution result** and must not be read as broad safety: the same seed, applied to the wider held-out cohort, gave false-abort 0.47 (Sec. 6, Fig. 5). It demonstrates that *material-matched* seeding worked on that single 2026 NbSe₂ grid, and reinforces the per-cohort-calibration requirement — not cross-cohort transport.

8 Live pilot: the abort machinery works end-to-end

Beyond retrospective replay, we ran the policy *live* in the AiiDA daemon through a staged, gated Fe-doped NbSe₂ pilot (Gate 0 $\rightarrow$ read-only pre-flight $\rightarrow$ advisory smokes $\rightarrow$ gated package install/-validation $\rightarrow$ stock baselines $\rightarrow$ ml-assist arm). Five systems, each paired against a matched stock counterfactual (same structure, cold start, mixing, segment cadence, partition). The pilot validates the *machinery*, not generalization; at $N=5$ every effect size is descriptive.

What the pilot shows. (i) A **true early abort** (S5) fired live end-to-end — hook $\rightarrow$ numpy inference $\rightarrow$ Mondrian per-stratum threshold $\rightarrow$ decision $\rightarrow$ telemetry $\rightarrow$ exit 701 — and saved ≈ 130 iterations. (ii) A **would-be false abort was correctly withheld** (S1, advisory), confirming the conservative-advisory default is the right guard for slowly-converging runs. (iii) An **honest miss** (S3): a run whose stall emerges late was not caught by the early window — consistent with the held-out finding that late stalls are the hard case. The abort mechanism, telemetry, and paired accounting are validated; the $N=5$ deltas are descriptive, not a powered savings claim.

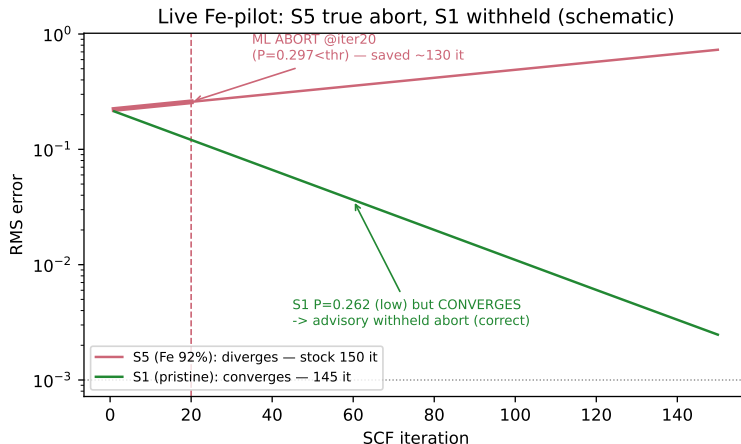


Figure 7: Live pilot exemplars (schematic of recorded runs). **S5** (Fe 92%) diverges; the frozen model scored $P(\text{converge}) = 0.297$ below the abort threshold and issued a live `ERROR_MLASSIST_ABORT` (exit 701) at cumulative iteration 20, saving ≈ 130 SCF iterations vs. its 150-iteration diverging counterfactual. **S1** (pristine) scored low ($P = 0.262$) yet converges; run advisory, its would-be abort was correctly *withheld* — the deliberate conservative trade-off.

9 The BdG Δ -channel

The standard parser exposes RMS and charge-neutrality but not the pairing amplitude. We wrote a parser for the `l-independent delta` lines KKRhost-BdG prints (two spin components per iteration, before each iteration marker), yielding a per-iteration Δ trajectory (length verified equal to the iteration count). Adding seven Δ -shape features lifts within-BdG *in-distribution* AUC from **0.946 to 0.957** (95% CI on the lift $[+0.000, +0.023]$; Δ -only AUC 0.884). (These are in-distribution 5-fold numbers, like Table 2, not held-out; and BdG runs must be scored with this BdG model — the normal model on BdG data reaches only AUC 0.58, Sec. 5.) The lift is modest here because the historical corpus barely contains pairing-*instability* runs (where Δ grows) — precisely the regime the parser targets for Vehicle A — so it is expected to under-state the value on the campaigns it is built for. The BdG model carries Δ ; every Vehicle-A run logs both the rms-only and rms+ Δ scores to measure the lift prospectively.

10 Workflow hardening: four BdG footguns and the memory number

A separate but essential outcome: making `kk_r_bdg_wc` run a multi-atom cell end-to-end. (Much historical BdG ran instead as standalone `KkrCalculations` in groups; the workchain is the path for *new* BdG runs, and the Δ -parser of Sec. 9 works on any BdG `out_kkr` regardless of how it was launched.) Four defaults, all tuned for a single-atom bulk reference, broke on an 8-atom NbSe₂ cell; each was fixed and validated, and the fixes are folded into the workchain (v0.2.1–v0.2.4) and mirrored to the working git branch:

1. **Multi-grid ranks.** MPI ranks must divide *every* step’s energy-point grid; a chain with a 24-point normal step and a 32-point BdG step requires a common divisor (or per-step ranks).

2. **Cell-aware RCLUSTZ.** The screening-cluster radius now inherits the cell’s value instead of a bulk default that overran the compiled cluster limit (NACLSD).
3. **NSPIN-aware spin coupling.** BdG with NSPIN=2 requires coupled spin channels; the workchain now drops the incompatible decoupling flag automatically.
4. **Cell-aware BZDIVIDE.** The k -mesh inherits the cell’s value; a bulk-dense mesh overran the compiled k -point limit (KPOIBZ) once BdG lowered the symmetry.

With these fixed, an 8-atom BdG chain completed for the first time, giving the **Vehicle-A memory number**: BdG job memory is $\approx 1.0 \text{ GB/rank}$, *flat with rank count* (energy-parallel replication), i.e. $2.32\times$ the same-cell normal-state memory — a mesh-robust scaling factor used for cluster packing (Fig. 8).

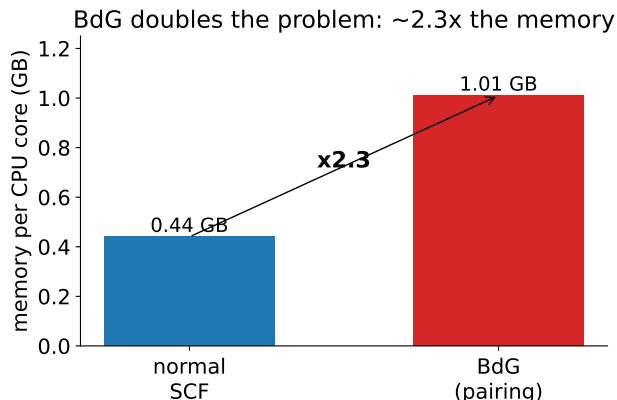


Figure 8: The BdG cost. Doubling the Hamiltonian into Nambu space raises memory per core by $\approx 2.3\times$; the factor is stable across core counts, which sets how many jobs fit on a node.

11 Software package

Everything ships as an installable `aiida-kkr-mlassist` extension (src-layout), with the pure-numpy core cleanly separated from the AiiDA workchain so inference needs no AiiDA and the daemon needs no scikit-learn:

```
aiida_kkr_mlassist/
  features.py canonical 15-D feature schema (train == infer)
  infer.py MLAssistModel - numpy random forest (sklearn-exact)
  conformal.py ConformalAbortPolicy + MondrianConformalAbortPolicy
  ranks.py multi-step MPI-rank / energy-grid validator
  bdg_parser.py BdG Delta-channel parser + features
  decision.py config, decision state-machine, telemetry, graceful failure
  workchain.py MLAssistKkrScfWorkChain (entry point kkr.mlassist)
  models/ frozen artifacts + manifests + MODEL_CARD.md
```

The workchain is opt-in (an absent input reproduces stock behaviour) and records every decision in node extras (advisory shadow data and audit trail). A unit-test suite (feature parity, golden parity, rank validation, conformal shadow-replay, Mondrian control, decision state-machine, BdG parser, dual-scoring) runs under the daemon interpreter to prove the deployment path is scikit-learn-free.

12 Pre-registration and success gates (superseded)

Deployment was originally pre-registered as two campaigns (one shared preamble) with frozen model hashes, the estimand definitions (TOTAL_CAMPAIGN_ITER, MEDIAN_ITER_TO_CONV), an A/B fairness rule (interleaved within a partition, wall-clock never pooled across partitions), a numeric conformal-violation trip-wire, and a pre-commitment that failures are reported as-is. **This original plan — in particular the Vehicle-B throughput campaign and its *G3* target — was superseded** and is retained here only for provenance:

Vehicle A (BdG, conservative). rms+ Δ model, dual-scoring telemetry, $\alpha = 0.05$, Mondrian thresholds; a numeric near-criticality criterion controls how many cells are abort-enabled so the safety rule cannot silently starve the recovery gate; *G2* target: recover $\geq 50\%$ of the stock arm’s own waste, zero conformal violations, physics-equality pass. (*Not run — BdG live campaign not launched.*)

Vehicle B (normal, throughput). $\alpha = 0.10$; *paired within-system* A/B (Wilcoxon on iterations) for power; a paired warm-and-cold subset; an explicit racing ladder, commit rule, and warm-start donor-compatibility checks; *G3* target: median iterations-to-converge and total campaign iterations both down $\geq 25\%$.

What actually happened. The full Vehicle-B campaign was *not* run: the planned dopant breadth (Mn/V/Co) lived on the CLAIX cluster, which became unavailable, collapsing the design to an Fe-only iffslurm pilot. That pilot ($N=5$, Sec. 8) validated the abort *machinery* but is underpowered for a Wilcoxon effect-size test, so **G3 is not an active claim**. The held-out benchmark (Sec. 5) then reshaped the deployment recommendation itself: per-cohort calibration with an RMS@10 comparator, rather than a fixed $\geq 25\%$ throughput gate. Any future campaign must re-pre-register against that recommendation.

Physics-equality tolerances (empirical). Set from the intrinsic numerical reproducibility — the per-run total-energy variation over the last converged iterations (plateau noise, median 1.6×10^{-7} Ry/atom over 2,833 runs). Between-run “rerun” scatter proved unusable (formula-level matching still conflates dopant configurations). The adopted total-energy tolerance is $3 \times$ the plateau-noise 90th percentile, $\approx 3 \times 10^{-5}$ Ry/atom — neither contaminated nor pathologically tight.

13 Side studies and honest negatives

Impurity (KKRimp) feasibility. Deferred: 2,390 impurity workchains exist, but their stored RMS trajectory has median length 1 — per-iteration RMS is not in provenance — so early-window prediction cannot be priced without richer logging. A logging change, not a modeling gap.

Warm-start (L3) matched pairs. Underpowered: zero system-matched cold/warm converged pairs exist historically, so the causal warm-start effect is not retrospectively estimable and is deferred to the pre-registered randomized design.

14 Reproducibility and governance

The AiiDA database is treated read-only for analysis; any code-environment or database change is confirmation-gated with a printed plan. Frozen models are versioned artifacts with SHA-256 and

a manifest recording feature schema, training-data hash, library versions, and out-of-sample AUC (Appendix B). The Replay-OOF Invariant, the two-denominator reporting rule, and the negatives-are-publishable pre-commitment are the standing methodological safeguards.

15 PotentialBank v1: L3 warm-start realized

The L3 lever — reuse a converged potential instead of a cold Voronoi start — is realized as **PotentialBank v1**: a provenance-clean catalog of certified single-structure converged KKR potentials, each carrying a byte-exact compatibility fingerprint. A donor is admitted only if it passes *E1 certification* verbatim: the parser succeeds and byte-exactly round-trips, the potential’s per-type *Z*-sequence matches the structure site order (empty spheres as $Z=0$), NATYP equals the site count, the cell is non-CPA and non-BdG, NSPIN/SOC/LMAX/mesh parse, no mapping is guessed, and the staged, AiiDA-retrieved, and manifest SHA-256 all agree. CPA, disordered, and BdG potentials are *excluded by construction*.

Coverage. v1 contains **238 certified donor nodes = 237 unique-SHA donors** (one recorded byte-duplicate: E1 Bi₂X₄ pk 24009≡24011), spanning **8 families, 35 materials, 42 compatibility classes**, of which **33 materials are F2-eligible** (≥ 2 independent certified donors): NbSe₂/Bi (E1, 160 donors), simple metals Nb/Al/Cu/Ag/Mo (24), heavy-SOC Au/W/Pt/Sb/Pb (24), magnetic Fe/Ni/Co (13), ordered intermetallics CuZn/Ni₃Al/FeAl (6), covalent empty-sphere Si/Ge (4), the metallic layered chalcogenide TiSe₂ (2), and ionic rocksalt MgO/NaCl (4). Each family required its own certified settings recipe (screening cluster RCLUSTZ, Chebyshev log panel R.LOG/NPAN, BZ mesh, SOC key routing, explicit empty-sphere mapping, and a mixing-path variation to obtain independent F2 pairs for deterministic insulator SCF).

Warm-start savings (same-structure). Injecting a certified donor as the start potential cuts iterations-to-converge by large, reproducible margins, *donor-choice-independent* (33/33 targets: identical iterations from either donor) and direction-symmetric where cold-comparable:

family	cold iters	warm iters	% saved
simple metals	66–102	4–7	91–94 %
heavy-SOC	122–136	2–3	~97 %
magnetic	131–136	5–6	~96 %
covalent Si/Ge	117–119	2–3	97–98 %
intermetallics	128–152	5–7	95–97 %
TiSe ₂ (layered)	1188–1354	≤ 100	~92 %
ionic MgO/NaCl	171–750	2–500	33–99 %

The magnitude is set by the target’s mixing accelerator (a Broyden target polishes a warm start in a few iterations; a pure-linear target still crawls, e.g. 500 vs. 750 for MgO/NaCl), so each row is a *matched same-mixing* warm-vs-cold comparison and we report per-family, never a single pooled headline. Contour caveats are recorded (TiSe₂ is coarse-converged only, as the fine contour destabilizes the zero-gap semimetal). Coverage, F2-eligibility, warm-vs-cold, savings distribution, the parked-hard-case map, and the provenance workflow are in Figs. 1–6 of the v1 synthesis package.

Limitations. PotentialBank v1 supports **same-structure reuse only**; it is *not* a cross-material transfer claim and *not* a guaranteed acceleration (donor≡target geometry, chemistry, and settings; savings depend on structure/mixing/contour). CPA/doped/disordered and BdG are out of scope.

Nine hard cases are parked with root causes (MoS₂ and NbS₂ NACLSD, WSe₂ SOC contour/EMIN instability, TaS₂ SOC-NaN, TiAl VERTEX3D, Bi₂Te₃/Bi₂Se₃ hard-convergence, CaF₂/LiF). The warm-start outputs are benchmarks and are *not* themselves certified as donors. For `ml_assist`, the early-abort mechanism is validated live but broad ML transfer is *not* proven — it does not beat a simple RMS@10 baseline out-of-sample and requires per-cohort recalibration.

15.1 Related work

Workflow and provenance. We build directly on AiiDA-KKR / JuKKR [1], which provide the KKR restart and `potential_overwrite` primitives our warm start uses. `aiida-mlip` [2] is a sibling AiiDA plugin that applies the same provenance discipline to machine-learned interatomic potentials, a different target. **Predicting quantities to accelerate SCF.** A growing body of work *predicts* an electronic-structure quantity to speed up or bypass self-consistency: ChargeE3Net [7] predicts the charge density and reports a 26.7% reduction in SCF iterations when initializing DFT on *unseen* materials; SALTED [8] learns 3D electron densities via density fitting; and DeepH / DeepH-E3 [9, 10] predict the DFT Hamiltonian directly. `mp-pyrho` [6] provides representation-independent charge-density regridding and an associated density database. **Representations and MLIP tooling.** Dscribe [3] supplies atomistic descriptors; DP-GEN [4] performs concurrent-learning generation of deep-potential models; `dftio` [5] standardizes DFT outputs for machine learning. **Where we differ.** These methods *learn to predict* a density, potential, or Hamiltonian and, in the best cases, transfer across chemistries. PotentialBank v1 makes **no prediction and no cross-material claim**: it *reuses an existing, certified potential for the identical structure*. The novelty is the certification and retrieval policy — integrity, compatibility, scope, and auditability — not a new physical model or a learned mapping. We are not aware of a verified KKR-specific tool that provides this certified same-structure reuse layer.

What is and is not novel. We do **not** claim to have invented KKR restart, potential overwriting, or SCF warm starting — those primitives exist in AiiDA-KKR and are standard practice. We do **not** predict potentials, densities, or Hamiltonians, and we make no cross-material transfer claim. What is novel is the **provenance-certified reuse layer**: automated donor discovery over the provenance graph; a strict, auditable eligibility policy (byte-exact integrity, exact structural compatibility, non-CPA / non-BdG scope, no guessed mapping); an optional convergence advisor with explicitly benchmarked limits; and honest same-structure warm-start benchmarks together with a documented catalog of hard cases and negative results.

16 Status and roadmap

Complete and validated: the framework and its pure-numpy artifact; the conformal (Mondrian) abort policy; the live pilot (abort mechanism validated end-to-end, Sec. 8); the leakage-clean held-out benchmark (Sec. 5); the BdG Δ -parser and frozen BdG models; the `aiida-kkr-mlassist` package with tests; the BdG workchain hardening and memory number; the physics-equality tolerance; the pre-registrations. **Key finding for deployment:** the learned model does not beat a simple RMS@10 rule out-of-sample on the primary cohort, does not transfer across materials, and its conformal guarantee requires *per-cohort recalibration*. Deployment is therefore per-cohort and baseline-benchmarked: the default scorer on the primary cohort is RMS@10, and any campaign must recalibrate the abort threshold on its own converged runs and compare model vs. RMS@10 before enabling abort. **Remaining modeling gap:** late-stall detection (largely missed by the early

window, for both the model and RMS@10). **L3 (warm start) is delivered** as PotentialBank v1 (Sec. 15): 237 certified single-structure donors across 8 families with measured same-structure savings; **L2** (mixing search) remains estimated by the pre-registered campaigns. PotentialBank v1 is same-structure reuse only — no cross-material transfer, no guaranteed acceleration.

17 Software and data availability

The software is open source under the **MIT** license and is available at https://github.com/MRHemmati/kkr_ml_tools (a tagged release and archival DOI will be minted on acceptance and recorded here). The repository ships the installable `aiida-kkr-mlassist` package and the `kkr_convergence_advisor`, unit tests, the frozen model artifacts with SHA-256 manifests, the PotentialBank v1 manifest / tables / figures, the advisor index, and two walkthrough notebooks. **Data availability:** the derived artifacts (manifests, benchmark tables, model manifests) are included; the raw KKR potential files are *not* redistributed — they remain in the AiiDA provenance store and are referenced by byte-exact SHA-256 and provenance handles (`seg_pk`, `wc_pk`, `uuid`), so their identity is verifiable by anyone with access to that store. Live KKR submissions require the JuKKR codes and the authors’ AiiDA database and cluster environment. *Versioning:* the Python *package* (`aiida-kkr-mlassist`) and the *data release* (PotentialBank and advisor index) are versioned independently — the package is at 0.1.1, while the certified potential catalog and advisor index are the frozen *PotentialBank v1* release; these are distinct artifacts and the different version labels are intentional, not an inconsistency.

18 Reproducibility

Offline, without HPC or AiiDA: install from source (`pip install -e ., numpy-only core`), load the advisor index (`kkr_convergence_advisor.load_index()`), run the self-asserting unit tests (`PYTHONPATH=src python tests/*.py`), and execute the two walkthrough notebooks (the submission tutorial is safe-by-default and submits nothing). A lightweight smoke test reproduces the headline counts (237/238/8/33); `sha256sum -c reports/potentialbank.v1/MANIFEST.sha256` verifies the synthesis bundle. The reviewer-run checklist gives exact commands and expected outputs.

CRedit author contribution statement

TODO before submission: complete CRediT roles per author (e.g. Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing — original draft, Writing — review & editing).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Use of AI assistance

AI coding and writing assistants were used to help draft text, generate code-edit suggestions, and organize reports; all scientific decisions, calculations, validation, and final claims were reviewed by the author.

A Result tables

Table 2: Per-material convergence AUC, *in-distribution* 5-fold cross-validation (deduplicated runs). Strong where data is plentiful; honestly weaker/underpowered where thin. **These are same-distribution CV numbers, not the deployment metric** — for the leakage-clean held-out result see Table 1. No cross-material transfer is claimed.

family	n	OOF AUC	95% CI
NbSe ₂	373	0.918	[0.86, 0.97]
Bi/NbBi	38	0.849	[0.69, 0.97]
mixed/unknown ²	1398	0.950	[0.92, 0.97]
pooled (normal, frozen model)	1830	0.944 ³	—

Table 3: The ML margin over baselines, *in-distribution only*. Out-of-sample the relevant baseline is RMS@10 (not the weak extrapolation heuristic), and it matches or beats the model per cohort (Table 1).

method	in-distribution AUC	held-out (primary cohort) AUC
physical extrapolation heuristic	0.549	—
logistic regression	0.826	—
RMS@10 rule (simple baseline)	—	0.897
random forest (<code>ml_assist</code>)	0.944	0.831

Table 4: Marginal vs. Mondrian conformal, false-abort rate (out-of-sample). The marginal threshold under-protects the flat-starter stratum; per-stratum (Mondrian) thresholds restore conditional control.

α	marginal (overall)	marginal (flat-starters)	Mondrian (flat-starters)
0.05	0.049	0.086	0.047
0.10	0.098	0.164	0.093

³In-distribution 5-fold CV; *not* the deployment headline. Out-of-sample on the primary NbSe₂-normal cohort the frozen model reaches AUC 0.83 and is beaten by RMS@10 0.90 (Table 1).

²“mixed/unknown” = runs whose chemical formula could not be recovered from provenance and so could not be assigned to the NbSe₂ or Bi/NbBi families; they are the largest, cleanest group and anchor the pooled model.

Table 5: Calibrated early-abort on the primary NbSe₂-normal cohort (held-out, per-cohort recalibration, weights frozen; nested-bootstrap split-conformal, iterations). Realized false-abort holds at target for both; **RMS@10 recovers more and catches more doomed runs than the model at every α .**

α	policy	realized false-abort	failed-run recall	wasted-iters saved
0.05	frozen ML	0.036	0.53	0.54
0.05	RMS@10	0.039	0.64	0.61
0.10	frozen ML	0.096	0.62	0.62
0.10	RMS@10	0.100	0.78	0.73
0.20	frozen ML	0.196	0.71	0.69
0.20	RMS@10	0.186	0.86	0.81

B Model provenance

Frozen artifacts (pure-numpy, golden-parity = 0 vs. scikit-learn, versioned by SHA-256):

model	role	features	in-dist. AUC ⁴
<code>mlassist_v1_normal_fixed10</code>	normal-state	15	0.944
<code>mlassist_v1_bdg_rmsdelta</code>	BdG primary	22	0.957
<code>mlassist_v1_bdg_rms</code>	BdG missing- Δ fallback	15	0.946

Each artifact ships a JSON manifest (feature names, training-data SHA-256, library versions, creation timestamp, golden-parity value) and a model card stating intended use, per-system scope, and the Replay-OOF Invariant.

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⁴In-distribution 5-fold CV. Held-out, the normal model scores 0.83 on its primary cohort (Table 1); weights were unchanged through all held-out and pilot evaluations.

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